Marco Ulise Tighiliu

ROBOTICS SYSTEMS • ADDITIVE MANUFACTURING • CAD

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I use the best tool for the job, if the best tool for the task at hand is not already in my toolset, I learn it while pursuing the task.

Personal Profile

age 23

A heartfelt engineer who strives to gain a full understanding of systems. I believe that the more you learn, the more you realize how much you don't know. I relish working in Mechatronics engineering, with a keen accent on Software Development, Robotics, 3D object design and manufacturing.

Deeply Relevant for the Job At Bondtech

MicroAPIgRESTion

GitHub

Python Library

- **Library** designed for easy **HTTP route handling** on **microcontrollers** like **ESP32**, **RP2040** variants with less than **21kB memory** available. Useful for creating **asynchronous REST APIs** and **serving websites**, enables microcontrollers to **make** their **resources accessible** to other devices efficiently.

3Dinbox

Bucharest, RO

Tech Lead / Additive Manufacturing Engineer & Mechanical Designer ***Full-time Job since November 2024***

- **Independently led the technical direction** of the company, serving as the primary authority on all aspects of **3D printing**, including 3D printer **service & repair**, **client consulting** in choosing the best tool for their manufacturing processes, **3D printing service**, **3D scanning**, and professional **technical consultation and training** on additive **manufacturing methods** and machines.
- Experienced with **Klipper, Marlin, RepRap** Firmware and editing/**writing system configs**.
- Serviced and repaired **customer 3D printers**, **troubleshooting hardware** and **software issues** to restore full functionality.
- Specialized in **Multi-material 3D printing** via various multi-nozzle systems like **IDEX**, **dual nozzles**:
  - **Materials that chemically bond like**: ASA + ABS-GF or PETG + TPU
  - **Mechanically bonded multi-materials**: Material combinations that do not adhere chemically but instead **interlock** through **mechanical design**.

Designed and **remade 3D models**, working with both **CAD-based parametric models** and **organic sculpted models** to fulfill client specifications and ensure optimal **print-ability**.
Some projects I worked on:

Bird repellent Capacitor	Assisted a client in **prototyping** a new version of an **electronic bird repellent** system designed to keep building rooftops clean by **temporarily disrupting a specific protein in birds' brains**, causing momentary blindness that makes them leave the area before regaining their sight. I provided **guidance on design** and **material selection**, and **manufactured** a **modular prototype** using **3D printing**, allowing for **scalable coverage** and **improved effectiveness** against larger birds. By using **PA6-GF** for its **insulation** properties, we **simulated** the **final product** while **significantly reducing costs**, **avoiding** a second **€10,000** investment **in injection molding** by **validating the design** beforehand.
Camera Drone Attachment	A Drone Shop wanted us to **Design** an **electronics enclosure** that conects a 3 Degree of Freedom **Programmable Camera** to a **Drone**, while compactly **routeing** the cables of the camera twoards the **microcontroller's** location, and **enclosing** the camera **electronics**, to **protect** the camera from **rain** and in case of **impact** while maintaining the **widest field of view** possible. I **managed everything independently**, from **project requirements gathering**, CAD-**design with revisions**, **material selection**, **manufacturing** and client **requirements validation**.
Utility Mobility Aid Kart	Assisted in developing a functional utility kart **prototype**, **designed** to **aid** individuals with **mobility impairments**. **Adapted** and **redesigned** the original model **for additive manufacturing**, and **fabricated** a scaled (1:2) **mechanical system with** multiple **moving parts** under **tight time constraints**.

Worked with a variety of **3D printers**, from **desktop FDM**, to **industrial FDM**, and **large-format** 3D printers.

Processed and **optimized** prints using a wide range of **materials**, **up to 350°C** on the nozzle, **PPS** and **Carbon-fiber**, **Glass-fiber composites**.

IoTFleet.js

GitHub

JavaScript Library

- Easy-to-use solution for **robotics applications** in which **multiple microcontroller** devices need to communicate seamlessly **over-air** with **non-blocking IO**.
- **Easy**-to-implement **host-side interface** for use in **multiple languages** and **contexts**.

Skills

Programming	C, C++, Python, Embedded (Microcontrollers, PLC, Single-board-computers), Computer vision, JavaScript, Web (HTML, CSS), REST API
3D Printing	Multi-material Printing, Optimising materials prone to warping (Unfilled Nylon, PC) for dimensional accuracy, Material selection
Robotics System	System Design, Communication Protocols (USB, UART, SPI, I2C, CAN Bus), Electric Circuit Design, System assembly, Electric fault diagnosis, Electronic Schematic Analysis, project requirements gathering and validation, repair & troubleshooting (specialising in 3D printers), prototyping and integration for Additive Manufacturing
CAD	SolidWorks, Onshape, FreeCAD
Collaboration	teamwork oriented, peer-to-peer value recognition, adaptable communication
Work ethic	self-driven, resilient, proactive problem-solving

Latest Education

Fontys School of Applied Sciences

Eindhoven, NL

Bachelor in Mechatronics *2021 - 2025*

- **Fontys Simulated Industrial Manufacturing**, a 4-quarter project featuring a **Robotic Arm vehicle** able to displace objects. We designed and manufactured a **3D printed Inverse-Kinematics Robot Arm** with free 360°+ rotation, commanded via **Bluetooth**, by running **C++** on various **Microcontroller** modules.
  Presented the project to a medium audience, showcasing my **presenting** skills by utilizing human movement as an easy to grasp example of inverse-Kinematics.
  Followed the industry-proven **V-model Systems Development Life-cycle**.
- **Modules: Robotics System Design, Embedded, IoT, Electrical Engineering, AI, 3D printing, Machining.**

Other experiences

Rainbow Bucharest

Bucharest, RO

Software Engineer & Mechanical Designer *Freelance projects in 2023*

- **Recovered 6000 lost** customer **emails** from a failure in the proprietary newsletter subscription solution, using **IMAP**
- **Replicated** custom **replacement parts not available** on the market using **CAD** software, including two **Motor Stator** models used in the **Rainbow D4 & eSeries**, intended for **production 3D printing.**

ZestOnScreenCapturer.py

GitHub

Python Library

- A high-frame-rate application frame capture solution designed to capture **live video feed** from applications in **Python** at **60 fps**. This **exceeds** the **25** frames per second offered by using basic **OpenCV**. Initially developed for smooth video game feed acquisition for **AI model** gameplay implementation and training.

Certified SOLIDWORKS Associate in Mechanical Design

Eindhoven, NL

Mathematics course writting assistant in LaTex

Fontys School of Applied Sciences

- Mechatronics First Year Math2B Course

Freelance projects in 2023